

Data-Driven Networks: IoT Analytics Revolution

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Analytics Foundations



Defining Network Analytics Today



01

Big Data Principles

Network analytics leverages big data principles to automate the ingestion and correlation of telemetry from various sources, replacing reactive approaches with data-driven governance.

02

Real-Time Interrogation

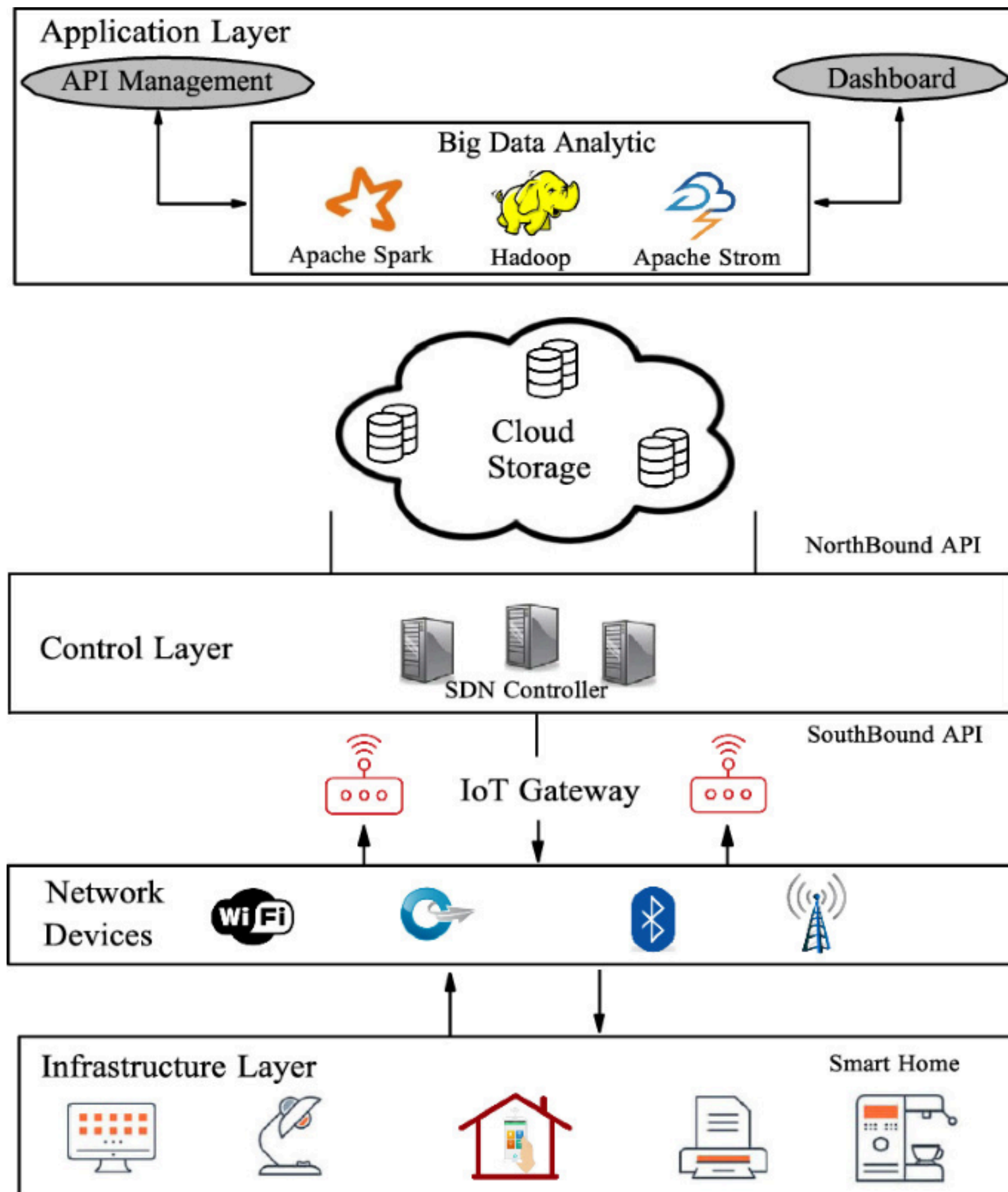
It enables real-time interrogation of multi-source data, including routers, switches, IoT sensors, and traffic flows, providing instant insights for network management.

03

Cloud Storage & AI Refinement

Data is stored in cloud-based systems and refined using AI algorithms, ensuring scalable and efficient processing for actionable insights.



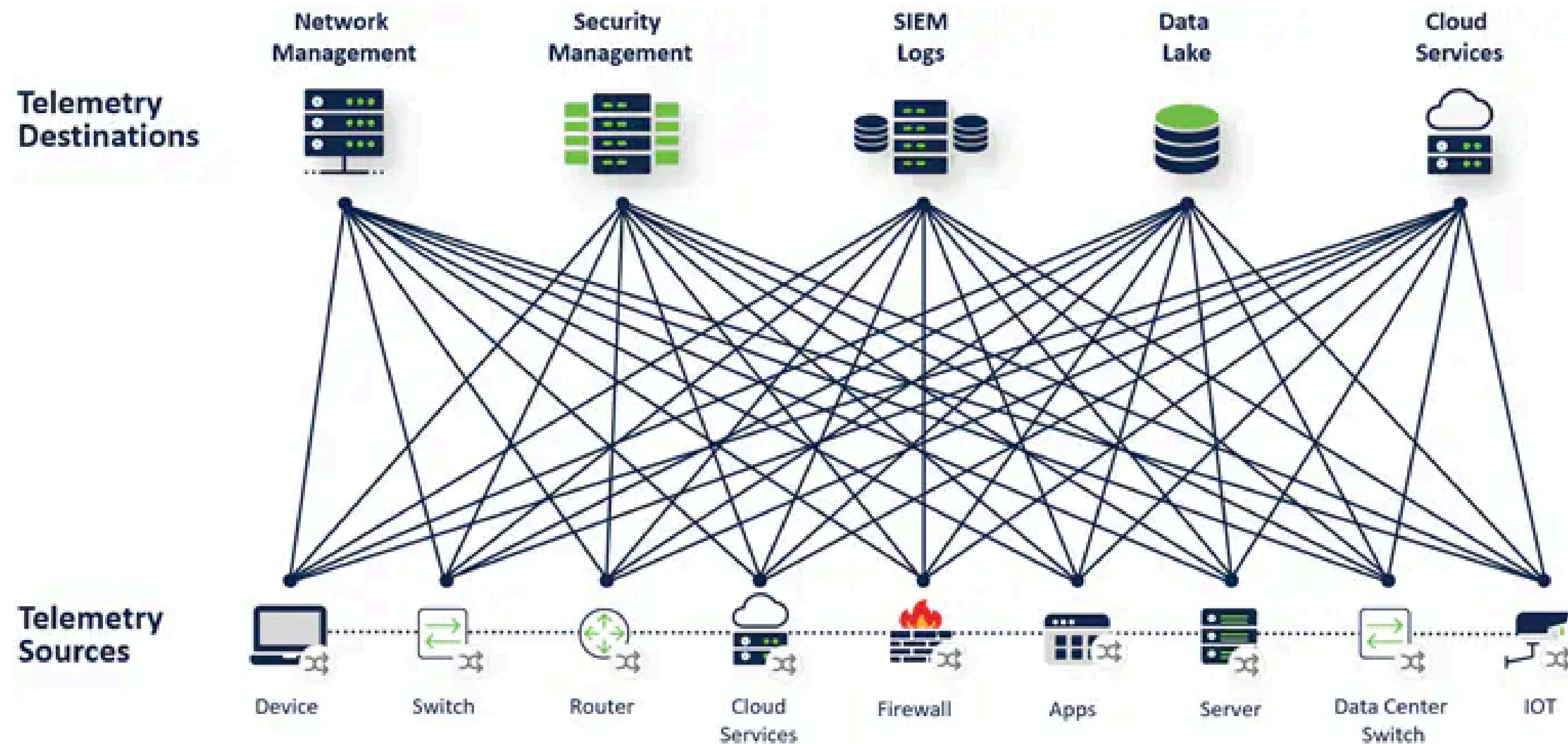


- Infrastructure Layer: Smart devices collect data.
- Network Layer: Wi-Fi, Bluetooth, etc., send data to gateways.
- Control Layer: SDN Controller manages data flow via IoT Gateways.
- Cloud Storage: Stores IoT data.
- Application Layer: Uses tools like Hadoop and Spark for analysis; results shown on dashboards.
- APIs: Southbound connects devices; Northbound connects cloud and apps.

Edge nodes enrich packets, and cloud pipelines aggregate these heterogeneous feeds through real-time telemetry and protocol polling, forming a unified, time-stamped data lake.

Telemetry Sources & Collection Paths

Diverse Telemetry Sources



02

Analytic Models



Four Lenses of Network Insight



01

Descriptive Analytics

Descriptive analytics quantify what happened by establishing network performance baselines and tracking device activity trends for historical context.

02

Diagnostic Analytics

Diagnostic analytics trace why anomalies emerged through root cause analysis, enabling precise identification of issues within complex networks.

03

Predictive Analytics

Predictive models forecast traffic surges, failure probabilities, and capacity needs, allowing proactive planning and resource allocation.

04

Prescriptive Analytics

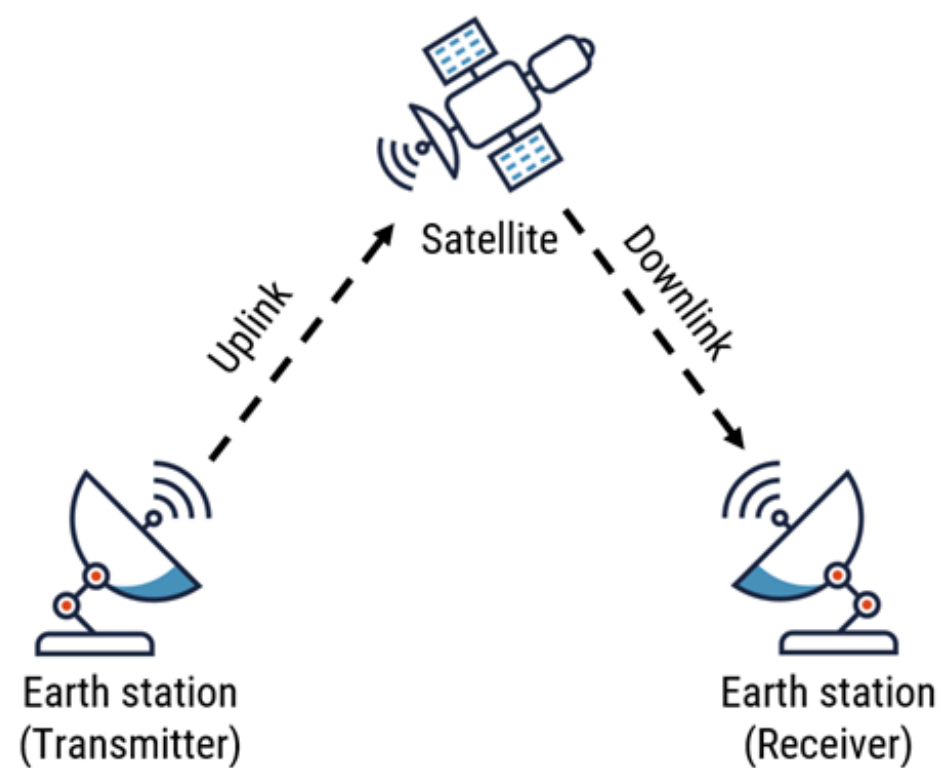
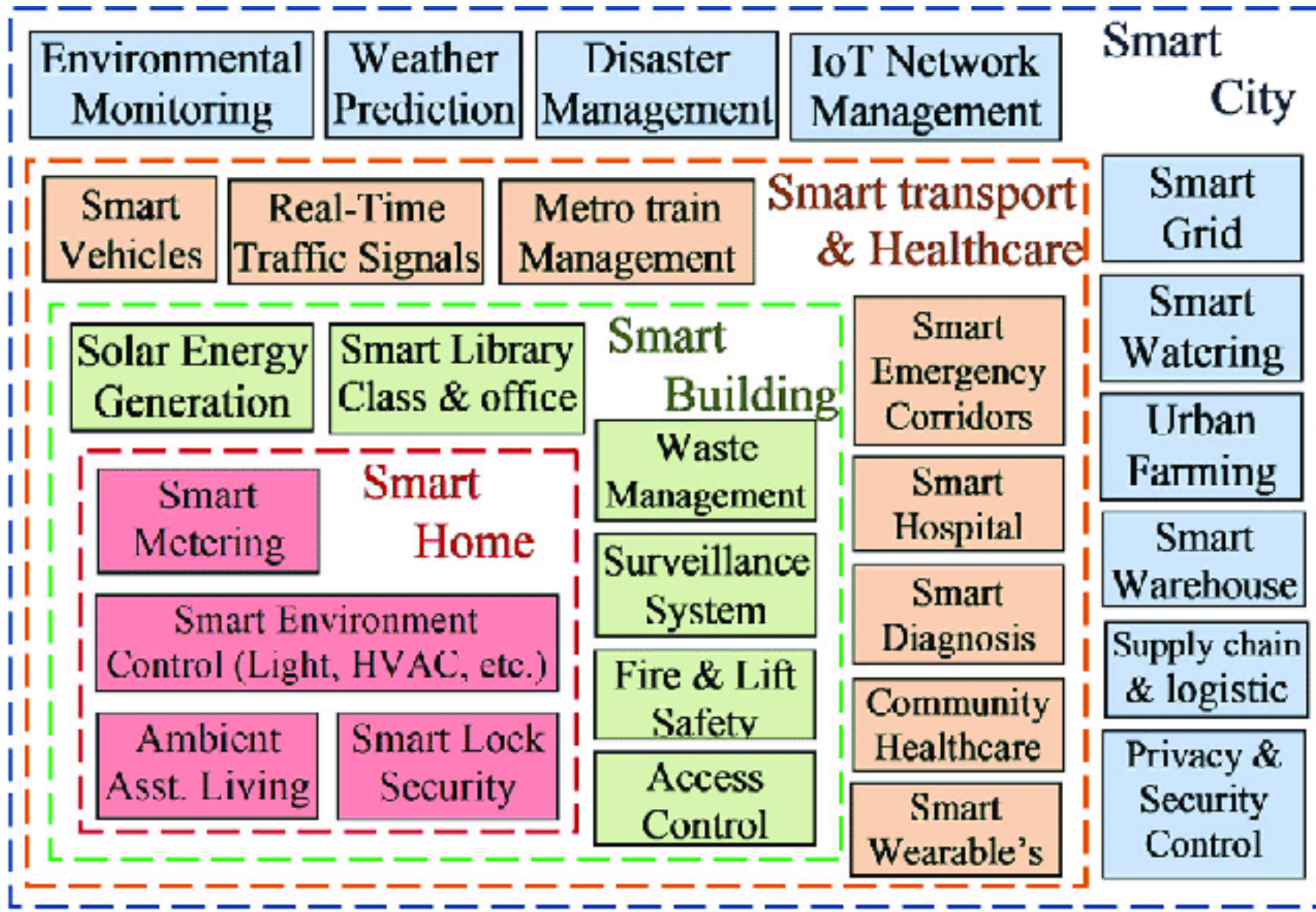
Prescriptive engines recommend or trigger automated remediation actions, optimizing network performance and ensuring continuous operation.

Managing IoT Scale

Millions of diverse IoT sensors emit structured and unstructured data, demanding real-time edge preprocessing and congestion-aware bandwidth dimensioning to maintain SLA compliance and prevent uplink saturation from aggressive or rogue endpoints.

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Service Level Agreements (SLAs)
 Best Practices

- 1. Create separate SLAs for each IT service you need to measure
- 2. Do not create SLAs that cover all your organization's divisions
- 3. Align SLAs with the customer's desired outcome
- 4. Make SLAs measurable
- 5. Review & adjust SLAs periodically
- 6. Ensure SLAs account for both usual & unusual exceptions



03

Security & Performance





Behavioral Threat Hunting

Baseline Tuning

By baseline-tuning entity behavior across encrypted and clear-text traffic, analytics can flag micro-anomalies and detect emerging threats.

Endpoint Profiling

Analytics auto-profile endpoints, accelerating incident response and enforcing adaptive security policies for dynamic threat containment.

Forensic Reconstruction

Forensic capabilities allow for detailed reconstruction of security incidents, providing valuable insights for future threat prevention and investigation.

Proactive Performance Engineering



Real-Time Correlation

Real-time correlation of device health, flow latency, and QoS metrics pinpoints bottlenecks and predicts saturation, enabling proactive performance management.



Automated Optimization

Automated tuning of routing and queuing parameters ensures optimal network performance, reducing downtime and maximizing resource utilization.

04

Architecture & Future



Scalable Analytics Stack



Cloud Data Lakes

Cloud data lakes on platforms like Hadoop or Spark store vast amounts of network data, providing scalable storage solutions for analytics.

Stream Processing

Stream processors like Kafka or Flink enable sub-second ingestion of data, ensuring real-time processing capabilities for dynamic networks.

Integration with NMS

RESTful APIs and interactive dashboards integrate seamlessly with legacy network management systems, delivering a unified view of network performance.

Self-Healing & Edge Evolution

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AI-Driven Autonomy

AI-driven closed loops will autonomously reconfigure faults, enabling self-healing capabilities that minimize downtime and improve network reliability.

Edge Analytics

Edge analytics will shrink latency for 5G and IIoT applications, providing faster insights and reducing the load on central data centers.



Digital Twins

Digital twin technology will simulate policy impacts, allowing for virtual testing and optimization of network configurations before deployment.

Federated Learning

Federated learning will secure multi-domain cooperation, enabling collaborative intelligence across distributed network environments.

05

Careers & Impact



Industry Use-Case Spotlight



Smart Cities

Smart cities balance traffic and energy via sensor fusion, optimizing urban infrastructure and improving quality of life for residents.

01

Remote Healthcare

Remote healthcare devices assure patient safety by continuously monitoring vital signs and alerting healthcare providers to anomalies.

02

Emerging Career Pathways

01

Network Data Scientists

Network data scientists decode packet behavior, extracting valuable insights from complex network data to drive performance improvements.

02

IoT Systems Architects

IoT systems architects design scalable telemetry solutions, ensuring seamless integration of devices and efficient data flow within networks.

03

Big Data Engineers

Big data engineers optimize streaming pipelines, ensuring efficient data processing and storage for real-time analytics in networking.



06

Takeaways



Key Lessons for Leaders

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Scalable Storage

Invest in scalable storage and stream processing solutions to handle the growing volume and velocity of network data.

Security Analytics

Prioritize security analytics to build trust and protect networks from emerging threats, ensuring data integrity and compliance.



02

04

01

Telemetry Governance

Start with clear telemetry governance to ensure data quality and consistency across network devices and IoT endpoints.

03

Domain Expertise

Pair domain expertise with ML tooling to develop effective analytics models that address specific network challenges.



Next Steps & Call to Action



Audit Data Sources

Audit current data sources for completeness, pilot predictive models on controlled segments, align with 5G and edge plans, upskill staff, and establish vendor-agnostic standards for a future-proof analytics journey.



THANK YOU



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